Lab 1 – Wireshark Installation and Basic Packet Analysis

Objective:

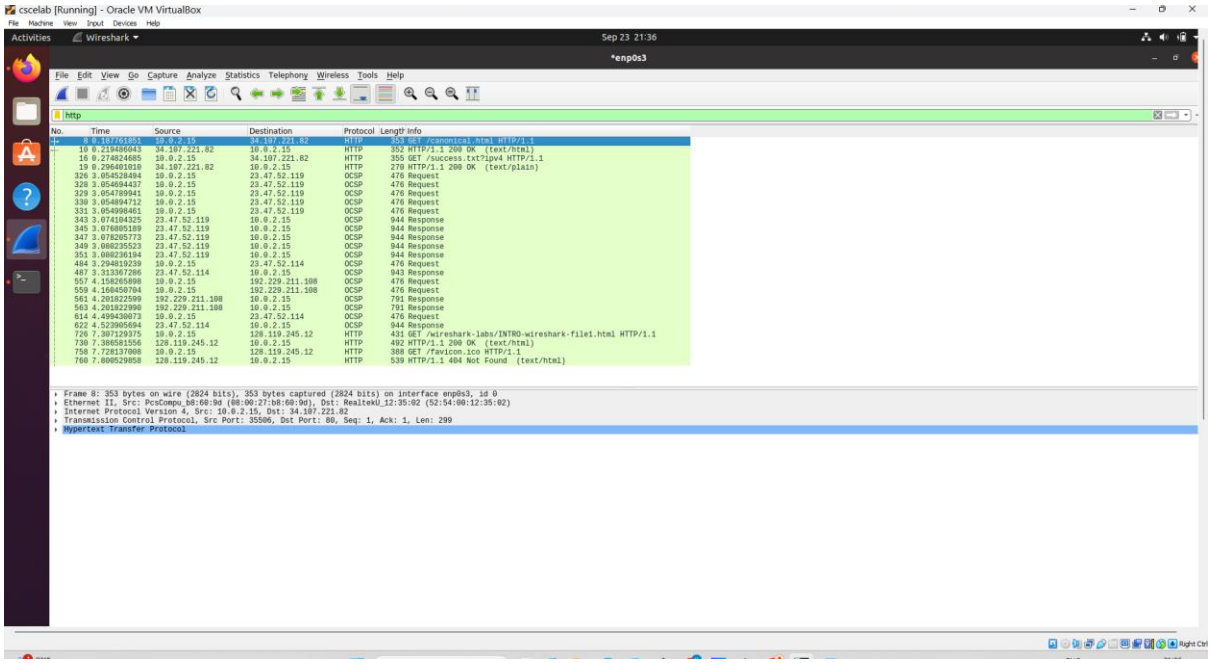
The purpose of this lab is to guide you through installing Wireshark and performing basic network traffic analysis. You will explore different protocols, analyze time differences between packets, and inspect IP addresses involved in HTTP requests.

Instructions:

1. Wireshark Installation:

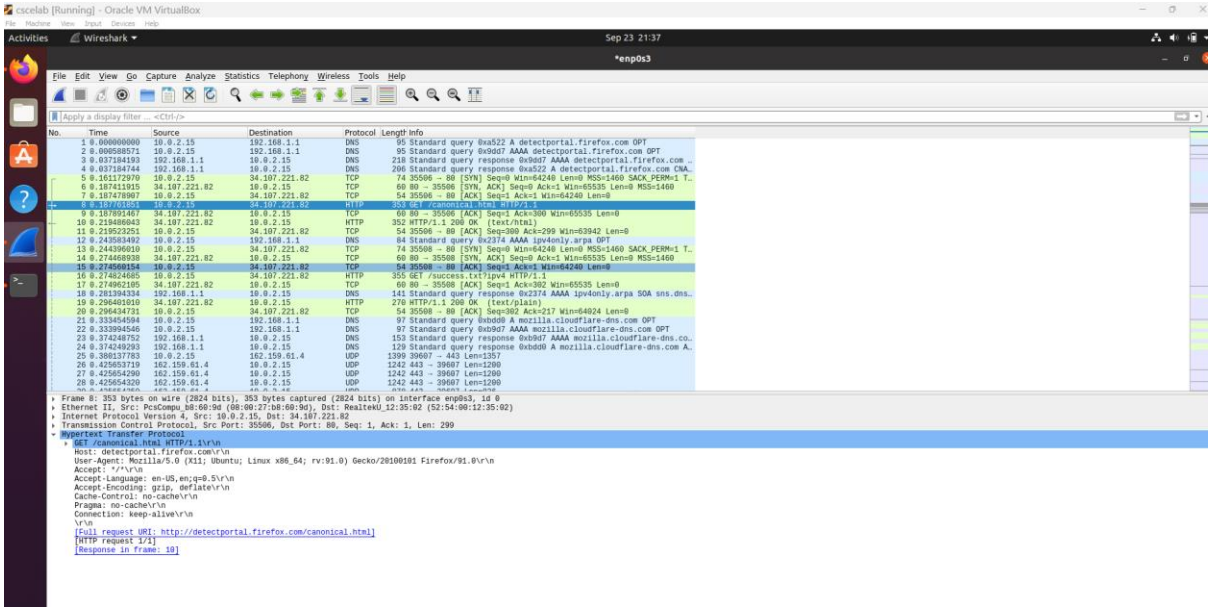
- Begin by installing Wireshark on your system. You can download it from the official Wireshark website (https://www.wireshark.org/download.html).
- Follow the installation instructions for your operating system.
- Once installed, open Wireshark and capture live network traffic.

We can see the 3 different protocols that appear in the protocol column in the unfiltered packet-listing window.



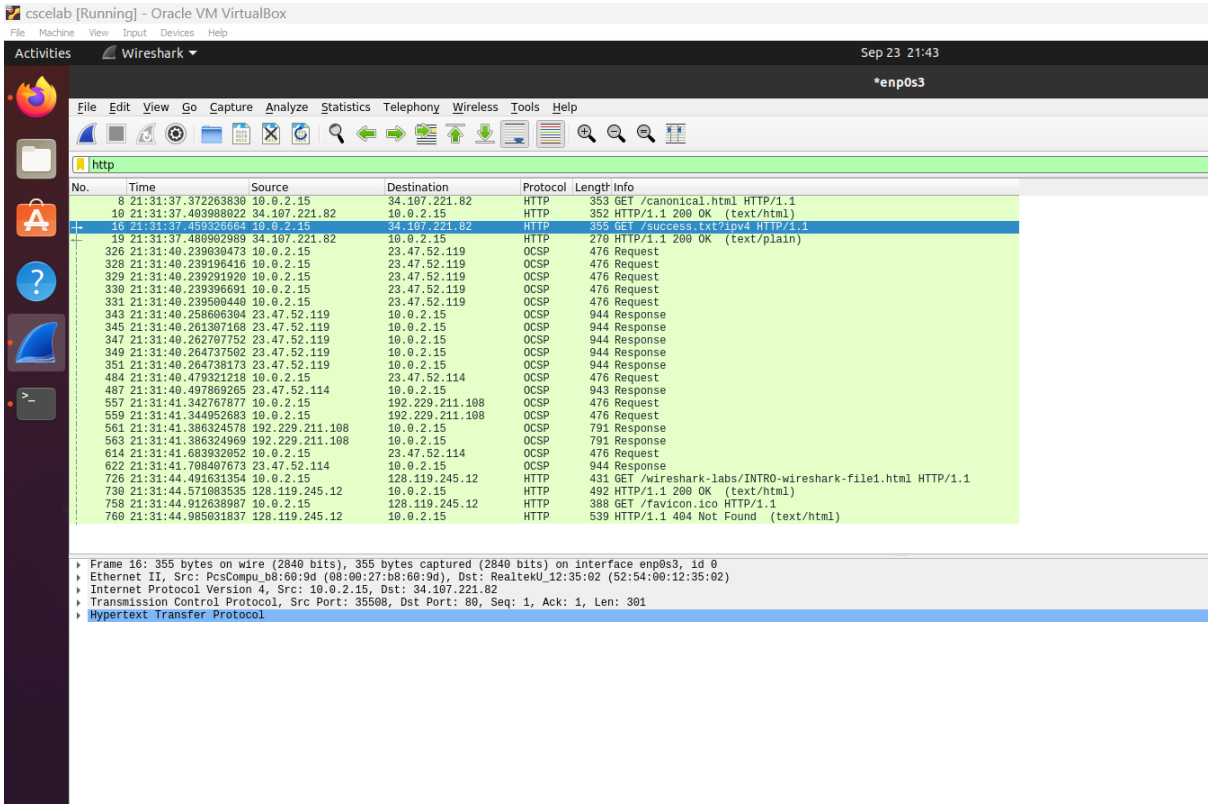
2. Protocol Analysis:

- Start capturing network traffic by selecting your network interface and clicking the "Start Capturing" button.
- In the unfiltered packet listing window, you will see various protocols listed in the "Protocol" column.
- **Task:** Identify at least three different protocols (e.g., HTTP, TCP, DNS) that appear in the packet listing.
- **Screenshot:** Take a screenshot showing the packet listing with the protocol column visible.
- **Question:** Briefly describe what each protocol represents and how it functions within the network.



3. Time Difference Between HTTP GET Requests:

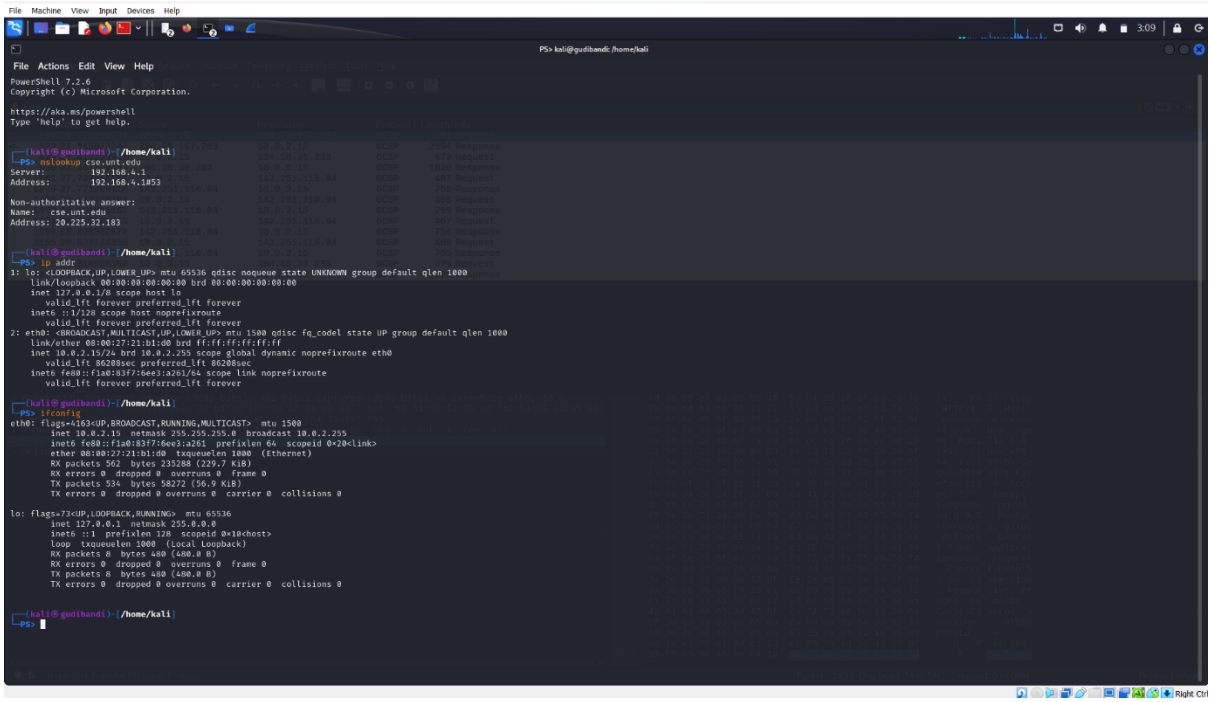
- Filter the capture for HTTP traffic by typing `http` in the Wireshark filter bar.
- Locate two HTTP `GET` requests in the packet listing.
- **Task:** Check the time difference between the two HTTP `GET` packets by selecting the first packet, right-clicking, and selecting "Set Time Reference (toggle)." Then, look at the second HTTP `GET` packet to see the difference.
- **Screenshot:** Take a screenshot showing the time difference between the two packets.
- **Question:** How many milliseconds passed between the two HTTP requests? Why might there be a delay between them?



We can check the two-time differences between two http get packets. There is a milliseconds difference between those packets.

4. IP Address Identification:

- Within the captured packets, locate the HTTP requests sent to `cse.unt.edu`.
- **Task:** Identify the IP addresses of both the `cse.unt.edu` server and your host system.
- **Screenshot:** Provide a screenshot showing the `cse.unt.edu` IP address and the IP address of your host system in the packet details pane.
- **Question:** What is the IP address of `cse.unt.edu`, and what is the IP address of your host system? Briefly describe the importance of IP addresses in network communication.



The above screenshot depicts the ip addresses of cse.unt.edu and address of host system.

Ip address of cse.unt.edu is 20.225.32.183

Ip address of host system is 10.0.2.15

5. HTTP Request and Response Analysis:

- Find both an HTTP `GET` request and its corresponding `OK` response in the capture.
- **Task:** Examine the details of the request and response packets, including source/destination IP addresses and key header information (e.g., `Host`, `User-Agent`, `Content-Type`).
- **Screenshot:** Provide a screenshot showing the details of both the `GET` request and the `OK` response.
- **Question:** What information can you gather from these packets about the client and server? Explain the significance of `GET` requests and `OK` responses in the HTTP protocol.

The above screenshot gives the information of both http GET and OK requests.

Submission:

- Submit the screenshots and answers to the questions above in a single document (PDF or Word).
- Ensure your answers are clear, concise, and demonstrate your understanding of the key concepts in network traffic analysis.